

Leveraging Physical Human-Robot Interaction for Surgical Robot Learning in Neuroendoscopy

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Abstract—Neuroendoscopy is a minimally invasive surgical approach with numerous benefits, but it presents technical challenges such as spatial constraints and the need for precise, repetitive movements. While surgical robots have the potential to address these challenges, their adoption in neuroendoscopy remains limited due to inadequate user interfaces and reliance on overly specific preprogrammed commands. This work introduces a robot learning algorithm that enables surgeons to kinesthetically train a robot arm to navigate and operate a neuroendoscope. Surgeons can manually guide the robot arm through task demonstrations, make real-time corrections, and teach the robot preferred surgical approaches, all within the surgical workspace constraints to ensure patient and surgeon safety. We aim to develop this machine learning-driven approach to create an intuitive shared control system between the surgeon and the robot, potentially reducing task completion times and enabling adaptation to anatomical variations.

Keywords—surgical robotics; physical human-robot interaction; learning from multi-modal feedback

I. INTRODUCTION

Neuroendoscopy is a minimally invasive surgical technique with numerous benefits, but it presents technical challenges such as spatial constraints and the need for precise, repetitive movements [1]. Surgical robots hold significant potential to automate repetitive surgical tasks with high precision, allowing surgeons to operate more efficiently in confined anatomical spaces [2], [3], [4]. However, surgical robots are yet to be widely adopted in neuroendoscopy, primarily due to their incompatible user interfaces [1], [2], [5], [6], [7]. As neuroendoscopy continues to expand the range of minimally invasive treatments for intracranial and skull base pathologies—such as endoscopic third ventriculostomy (ETV) for hydrocephalus and endoscopic endonasal approaches (EEA) for pituitary tumors—there is an increasing need to adopt surgical robots for neuroendoscopic applications.

In prior work, we conducted a systematic review of all surgical robots cleared by the United States Food and Drug Administration and found that robotic assistance in neuroendoscopy is currently limited to passive support of surgeon actions [7]. This slow progress is largely due to traditional control methods that rely on preprogrammed commands and still struggle with generalization despite extensive data collection, annotation, and prolonged offline training [8], [9], [10], [11]. To address these limitations, we are developing a robot learning algorithm using the Franka

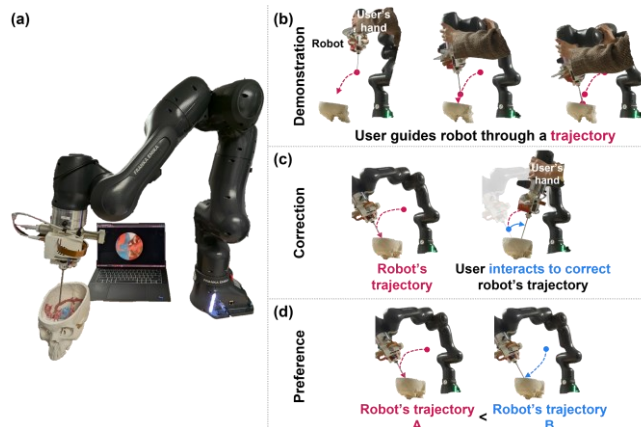


Fig. 1. (a) Our robot with a custom neuroendoscopy end effector. Our algorithm enables surgeons to teach the robot using (b) kinesthetic demonstrations, (c) corrections, and (d) preferences.

Research 3 robot (Franka Robotics, Germany) that enables surgeons to kinesthetically teach a robot arm to navigate and operate a neuroendoscope in anatomical models specific to ETV and EEA. Surgeons can manually guide the robot arm through task demonstrations, make real-time adjustments during tasks, and teach the robot their preferred surgical approaches, all within the constraints of the surgical workspace. We aim to develop this machine learning-driven approach to create an intuitive shared control system between the surgeon and the robot, potentially reducing task completion times and allowing for adaptability to varied patient anatomies. We have developed our algorithm in computer simulations of a virtual robot arm. We are subsequently implementing our algorithm in a physical robot arm equipped with a custom end effector designed for neuroendoscopy.

Key Research Questions. Our research aims to translate advances in human-robot interaction from assistive robotics to surgical robotics, enabling surgeons to intuitively teach a robot surgical tasks through physical interactions. Specifically, our work addresses the following questions:

- **RQ 1:** How effectively does our algorithm perform in a real robot with surgeon physical interactions, using only one anatomical model for validation?
- **RQ 2:** How effectively does our algorithm perform across anatomical variations and surgical conditions, using multiple anatomical models for validation?

II. BACKGROUND AND RELATED WORK

Uniting Physical Interactions. Physical human-robot interaction can be used to teach a robot desired behavior [12], [13], [14]. Building on the concept of shared control, a robot may perform tasks near a human end-user, who may physically interact with the robot. The human may first kinesthetically guide the robot arm through a full demonstration of a task [14], [15], [16]. Next, the robot attempts to perform the task autonomously under the observation of the human, who may push, pull, or twist the robot to correct the robot's motion [12], [13], [14]. Finally, as the robot gets better at understanding and performing the task on its own, the human can transition to observing and ranking the robot's behavior based on their preferences [14], [15], [17]. Each interaction modality provides distinct yet complementary information to the robot. While these physical human-robot interactions have been studied in assistive robotics, their potential remains largely untapped in surgical robotics, where most systems have traditionally relied on a surgeon-at-a-console paradigm [7], [18]. Our research explores how close physical interaction can facilitate safe and intuitive surgeon-robot collaboration, especially as surgical systems evolve toward modular, bed-mounted designs and incorporate higher levels of autonomy.

Learning Rewards from Multiple Types of Feedback. Recent end-to-end frameworks in assistive robotics integrate learning from demonstrations, corrections, and preferences by first learning a task reward model from scratch, and then optimizing this model to generate robot trajectories [14]. Since these approaches enable robots to learn new behaviors, it is important to ensure that these behaviors are safe in shared human-robot spaces and surgical workspaces. Our research focuses on augmenting existing methods with virtual constraints tailored to neuroendoscopy, such as workspace virtual walls and virtual remote center of motion constraints [19], [20]. This work also incorporates insights from user studies with surgeons performing surgical tasks for real-world validation.

III. PRIOR WORK

Algorithm Development. We have implemented an algorithm for unified learning from physical human-robot interactions, building on prior work [14]. The robot first learns a task reward model, represented as an ensemble of independently trained fully connected neural networks. This reward model maps robot states—such as the end effector pose or joint positions—to scalar value scores, and is trained to assign higher scores to states from human input trajectories than to states from nearby noisy alternative trajectories. After each interaction, the robot retrains its reward model. In the second part of the algorithm, the reward model is converted into a robot trajectory using constrained optimization. To tailor the algorithm for neuroendoscopy, we augmented the constraints with workspace safety limits—virtual walls and virtual remote center of motion constraints—and speed thresholds [19], [20]. We implemented our algorithm in custom PyBullet physics simulation environments with a virtual Franka Research 3 robot performing the task of positioning its end-effector at a target position on human anatomical models [21]. We evaluated learning using our algorithm compared to behavior cloning with metrics such as regret and root mean squared error.

Robot and Hardware Development. We have equipped the Franka Research 3 robot with a custom end effector compatible with rigid neuroendoscopes used in EEA and ETV procedures (ChannelScope and NeuroPEN, Medtronic/Clarus) [22], a single-board computer, and camera modules. This end effector enables the capture and near real-time streaming of endoscopic video synchronized with depth data. Additionally, we have implemented custom controllers and user interfaces to allow kinesthetic demonstration and correction recording and preference inputs on the Franka Research 3. We have also developed protocols to segment and construct 3D models of human nasal anatomy and brain and ventricular anatomy from CT and MRI scans.

IV. RESEARCH APPROACH AND FUTURE WORK

We will conduct a series of user studies to validate our algorithm in a real robot and gather end-user feedback to refine parameters and constraints. Participants will include neurosurgeons and ear, nose, and throat (ENT) surgeons familiar with endoscopy. We will ask participants to teach the robot to focus the endoscope on targets in anatomical models for ETV and EEA using different algorithms. Our algorithm performance will be benchmarked against baseline methods like behavior cloning and teleoperation. For our algorithm and behavior cloning, each participant will teach the robot using a fixed total number of interactions. For teleoperation, participants will operate the robot throughout the task using input devices such as a gamepad and keyboard. Each participant will experience teaching the robot using all algorithms in a randomized order, allowing us to compare performance. A 7-point Likert scale survey will follow each algorithm to capture feedback on ease of interaction, task accuracy, and preference for the method in ETV and EEA procedures. In addition, we will compare clinically relevant metrics across algorithms such as task completion time and motion smoothness [23].

In the first phase of the user studies, we will focus on RQ 1 by asking participants to teach their task in only one anatomical model using the different algorithms. For each learning algorithm, the weights of the reward model will be randomly initialized at the start of each trial such that the robot is learning the task from scratch. In the second phase, we will investigate RQ 2 and explore ways to assess and build on our algorithm to improve its performance across anatomical variations and surgical conditions. Participants will be asked to teach their task in different anatomical models that reflect common variations relevant to EEA and ETV [24], [25], [26] and realistic surgical challenges such as simulated bleeding [27], [28], [29]. We aim to investigate methods to accelerate training with our algorithm, such as exploring the potential for pre-training a reward model on similar tasks across different anatomical models, rather than restarting learning from scratch. Additionally, we will explore ways to augment our current reward model architecture to incorporate visual feedback from the sensors on our custom end effector.

ACKNOWLEDGMENTS

This work is supported by the Medical Scientist Training Program at the Icahn School of Medicine at Mount Sinai (NIH T32GM146636).

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